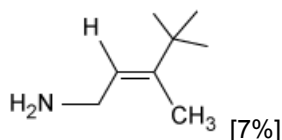
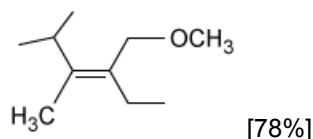


Which of the following compounds is most likely to be formed from an elimination reaction that produces a Z alkene?

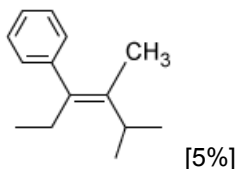
☐ A.



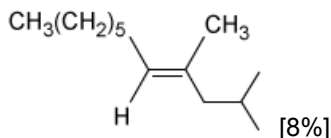
✓ ☒ B.



☐ C.



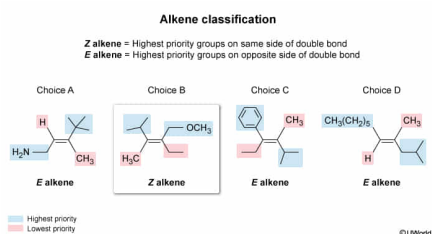
☐ D.



Correct

77% answered correctly

Explanation:



Alkenes belong to a class of diastereomers known as **geometric isomers**, and the substituents on these molecules differ in their arrangement about a double bond. Double bonds can be **di-, tri-, or tetra-substituted** and are classified as either **E/Z** or **cis/trans**. Classification is assigned based on the **Cahn-Ingold-Prelog priority rules**. If the **higher-priority groups** are on the **same side** of the double bond, the alkene is **Z**, whereas it is **E** when the **higher-priority groups** are on **opposite sides** of the double bond.

On the left side of the double bond in Choice B, the **isopropyl group** has higher priority than the methyl group ($-\text{CH}_3$) because the isopropyl carbon has three bonds to carbon and one bond to hydrogen whereas $-\text{CH}_3$ has three bonds to hydrogen. Because carbon has a greater atomic number than hydrogen, it is higher priority. The carbon on the right side of the double bond is bonded to two $-\text{CH}_2$ groups, so the next atom in each chain must be examined. The $-\text{CH}_2$ bonded to $-\text{OCH}_3$ is higher priority than the one bonded to $-\text{CH}_3$ because oxygen has a higher atomic number than carbon. Because the higher-priority groups are on the same side (top) of the double bond, the alkene is classified as Z.

Therefore, the structure for **Choice B** is most likely to form from an elimination reaction that produces a Z alkene.

(Choices A, C, and D) Each of these structures has the higher-priority groups on opposite sides of the double bond (one above, one below). Therefore, these alkenes are **E** and would not be products in a reaction that forms a Z alkene.

Educational objective:

Alkenes are geometric isomers that differ in the substituent arrangement about a double bond and can be classified as either **E** or **Z**. When the higher-priority substituents are on the same side of the double bond (top or bottom), the alkene is **Z**, and it is **E** when the higher-priority substituents are on opposite sides of the double bond.